

Applied Digital Logic Design (ECE101)

Department of E&CE

N.M.A.M. Institute of Technology, Nitte



UNIT – II

Simplification of Boolean Expressions :

Simplification of Boolean Functions using Boolean Algebra, Karnaugh Maps, Quine-McCluskey (up to four variables), Realization of Boolean functions using Basic Gates

06 Hrs

In many digital circuits and practical problems, we need to find expressions with minimum variables.

We can minimize Boolean expressions of 3, 4 variables very easily using K-map without using any Boolean algebra theorems.

K-map can take two forms:

1. Sum of product (SOP)
2. Product of Sum (POS)

SOP FORM

1. K-map of 3 variables

		BC			
		$B'C'$	$B'C$	BC	BC'
		00	01	11	10
A	A' 0	$A'B'C'$ 0	$A'B'C$ 1	$A'BC$ 3	$A'BC'$ 2
	A 1	$AB'C'$ 4	$AB'C$ 5	ABC 7	ABC' 6

SOP(MINTERMS)

8 Blocks = 1

4 Blocks = 1 variable term

2 Blocks = 2 variable term

1 Block = 3 variable term

2. K-map for 4 variables

		CD	C'D'	C'D	CD	CD'
		00	01	11	10	
AB						
A'B' 00	A'B'C'D'	A'B'C'D	A'B'CD	A'B'CD'		
A'B 01	A'BC'D'	A'BC'D	A'BCD	A'BCD'		
AB 11	ABC'D'	ABC'D	ABCD	ABCD'		
AB' 10	AB'C'D'	AB'C'D	AB'CD	AB'CD'		

SOP(MINTERMS)

16 Blocks = 1

8 Blocks = 1 variable term

4 Blocks = 2 variable term

2 Blocks = 3 variable term

1 Block = 4 variable term

Steps to Solve Expression using K-map

1. Select the K-map according to the number of variables.
2. Identify minterms or maxterms as given in the problem.
3. For SOP put 1's in blocks of K-map respective to the minterms (0's elsewhere).
4. For POS put 0's in blocks of K-map respective to the maxterms (1's elsewhere).
5. Make rectangular groups containing total terms in power of two like 16, 8, 4, 2, 1 and try to cover as many elements as you can in one group.
6. From the groups made in step 5 find the product terms and sum them up for SOP form.

1. We need to group the cells with a value of 1 and may not include any cell containing zero.
2. Groups may be horizontal, vertical, but not diagonal.
3. Groups must contain 1, 2, 4, 8, or in general 2^n cells.
4. The Group should be as large as possible.
5. Groups can overlap with each other.
6. Opposite or corner cells can be grouped.
7. Groups may wrap around the table. The leftmost cell in a row may be grouped with the rightmost cell, and the top cell in a column may be grouped with the bottom cell.
8. There should be as few groups as possible, as long as this does not contradict any of the previous rules.

K-map uses some rules for the simplification of Boolean expressions by combining together adjacent cells into single term.

Rule 1 – Any cell containing a zero cannot be grouped.

✗

		BC			
	A	00	01	11	10
0		1	0	1	0
1		0	1	1	1

Rule 2 – Groups must contain 2^n cells (n starting from 1).

✗

		BC			
	A	00	01	11	10
0		1	0	1	0
1		0	1	1	1

Rule 3 – Grouping must be horizontal or vertical, but must not be diagonal.

BC

A

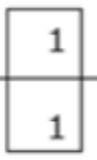
	00	01	11	10
0	1	1	1	0
1	0	0	1	1




BC

A

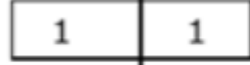
	00	01	11	10
0	1	1	1	0
1	0	0	1	1




BC

A

	00	01	11	10
0	1	1	1	0
1	0	0	1	1




Rule 4 – Groups must be covered as largely as possible.

BC

A

	00	01	11	10
0	1	0	1	0
1	1	1	1	1

×

BC

A

	00	01	11	10
0	1	0	1	0
1	1	1	1	1

✓

Rule 5 – If 1 of any cell cannot be grouped with any other cell, it will act as a group itself.

		BC			
		00	01	11	10
A	0	1	0	1	0
	1	0	1	0	1

Rule 6 – Groups may overlap but there should be as few groups as possible.

		BC			
		00	01	11	10
A	0	0	0	1	1
	1	1	1	1	1

Rule 7 – The leftmost cell/cells can be grouped with the rightmost cell/cells and the topmost cell/cells can be grouped with the bottommost cell/cells.

		BC			
		00	01	11	10
A	0	1	0	0	1
	1	1	0	0	1

Minimize the following Boolean expression using K-map –

$$F(A, B, C) = A'BC + A'BC' + AB'C' + AB'C$$

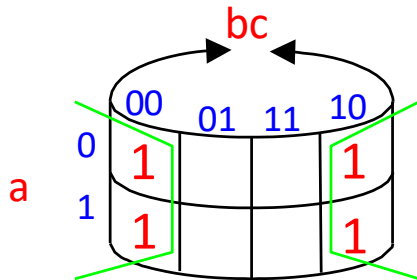
		BC			
		00	01	11	10
A	0	0	0	1	1
	1	1	1	0	0

		BC			
		00	01	11	10
A	0	0	0	1	1
	1	1	1	0	0

We have got two groups which are termed as $A'B$ and AB' . Hence, $F(A, B, C) = A'B + AB' = A \oplus B$. It is the minimized form.

In reality, the K-map is really a sphere (or can be visualized as a cylinder in some aspects) which has been flattened into a two-dimensional plane.





This figure demonstrates that the (10) column (right edge) is in reality adjacent to the (00) column (left edge) even though they don't look like they are adjacent.

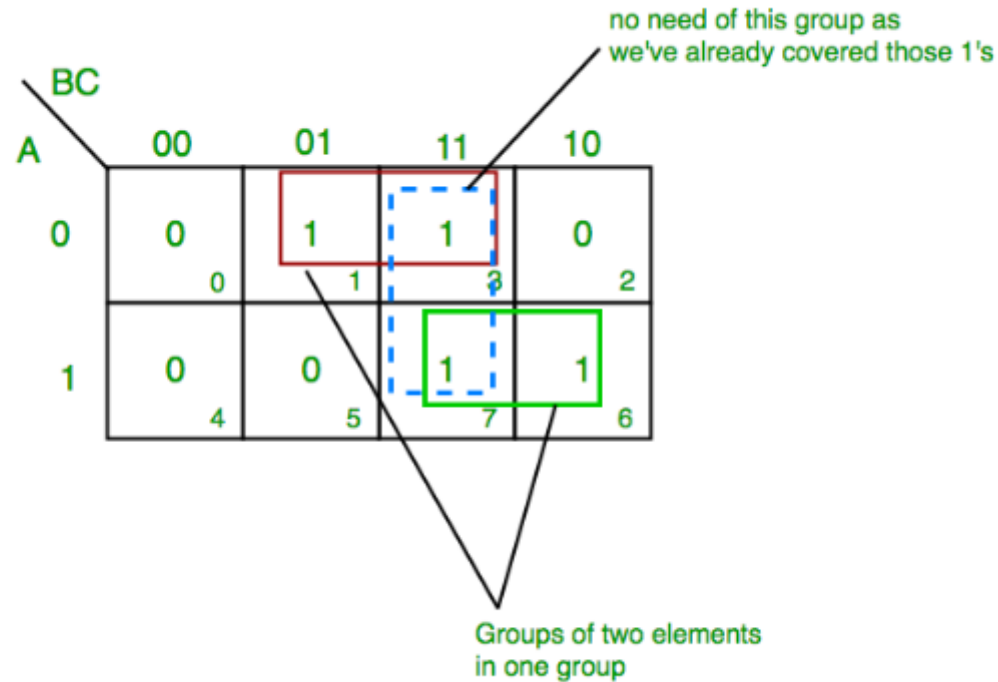
$$f(a, b, c) = \bar{c}$$

Note that between the columns (00) and (10) only the 'b' variable changes (which is the K-map requirement for adjacency.)

Now, due to this, the four 1's in the map can be grouped into a single group of four where the variable 'a' falls out of the expression vertically and the variable 'b' falls out horizontally.

The only variable which is left is the variable 'c' which is a '0' in those two columns so the final expression is a $\bar{c}$.

$$F = \sum m(1, 3, 6, 7)$$

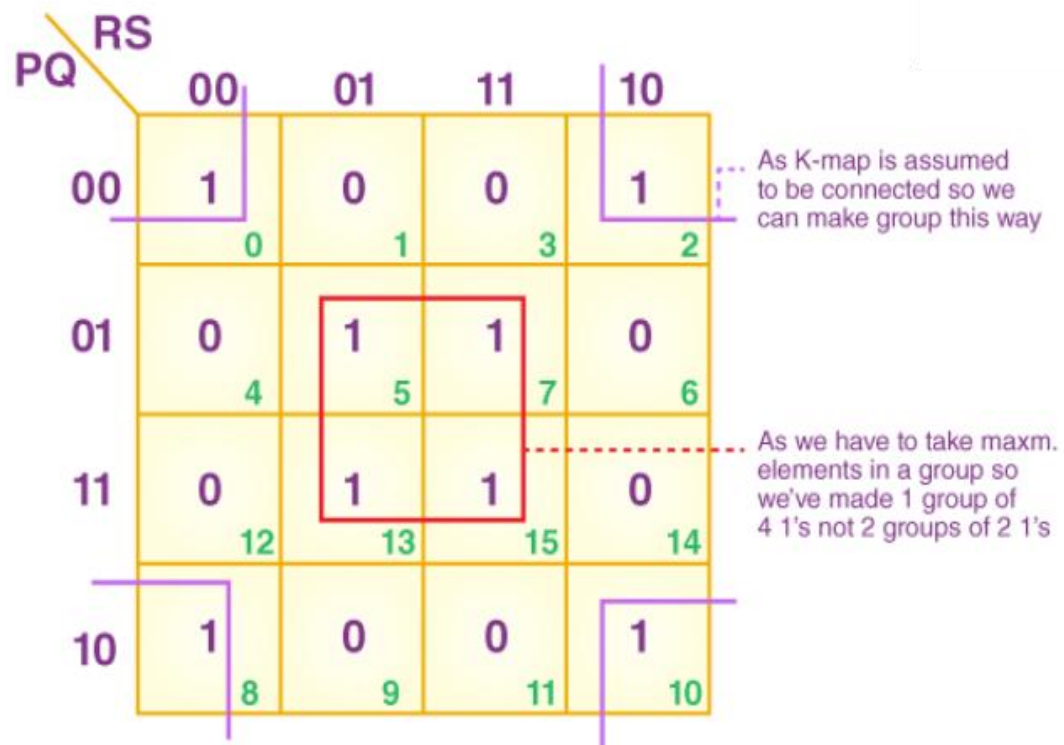


$\bar{A}C$

AB

$$F = \bar{A}C + AB$$

$$F(P, Q, R, S) = \Sigma(0, 2, 5, 7, 8, 10, 13, 15)$$



QS

$\bar{Q}\bar{S}$

$$F = QS + \bar{Q}\bar{S}$$

$$f(A, B, C, D) = \sum m(0, 1, 3, 4, 5, 7, 12, 13, 15)$$

A \ B		CD			
		00	01	11	10
00	1	1	1	0	
01	1	1	1	0	
11	1	1	1	0	
10	0	0	0	0	

A \ B		CD			
		00	01	11	10
00	1	1	1	0	
01	1	1	1	0	
11	1	1	1	0	
10	0	0	0	0	

$$f(A, B, C, D) = \overline{A}\overline{C} + \overline{A}D + B\overline{C} + BD$$

In Boolean algebra, a Boolean function or logic function whose output values are known for a subset of its inputs and has output which is a don't care condition for at least one of its input combinations is called an **incompletely specified function**.

Hence, in an incompletely specified function, the value of the function for don't care combinations can be selected either a 0 or a 1 depending on which one leads to the more minimal solution of the function.

For these combinations, the Karnaugh map (K-Map) entry is marked as X or dc, which indicates that the particular cell can be taken as a 0 or a 1.

$$F(A,B,C) = \sum m(1,2,7) + d(0,3,4)$$

BC		00	01	11	10
A	0	X	1	X	1
	1	X	0	1	0

0 1 3 2
4 5 7 6

Karnaugh Map

$$F(A,B,C) = \bar{A} + BC$$

Design a digital system whose output is defined as logically low if the 4-bit input binary number is a multiple of 3; otherwise, the output will be logically high. The output is defined if and only if the input binary number is greater than 2.

Truth Table

Inputs				Decimal Equivalent	Output
A	B	C	D		Y
0	0	0	0	0	X
0	0	0	1	1	X
0	0	1	0	2	X
0	0	1	1	3	0
0	1	0	0	4	1
0	1	0	1	5	1
0	1	1	0	6	0
0	1	1	1	7	1
1	0	0	0	8	1
1	0	0	1	9	0
1	0	1	0	10	1
1	0	1	1	11	1
1	1	0	0	12	0
1	1	0	1	13	1
1	1	1	0	14	1
1	1	1	1	15	0

AB \ CD	CD			
	00	01	11	10
00	X ⁰	X ¹	0 ³	X ²
01	1 ⁴	1 ⁵	1 ⁷	0 ⁶
11	0 ¹²	1 ¹³	0 ¹⁵	1 ¹⁴
10	1 ⁸	0 ⁹	1 ¹¹	1 ¹⁰

where X indicates Don't Care Condition

		CD			
		00	01	11	10
AB	00	X	X	0	X
	01	1	1	1	0
	11	0	1	0	1
	10	1	0	1	1

$$Y = \bar{B}\bar{D} + \bar{A}\bar{C} + \bar{A}BD + B\bar{C}D + A\bar{B}C + AC\bar{D}$$

BCD-to-Excess-3 Code Converter (1/2)

- Truth table:

	BCD				Excess-3			
	A	B	C	D	W	X	Y	Z
0	0	0	0	0	0	0	1	1
1	0	0	0	1	0	1	0	0
2	0	0	1	0	0	1	0	1
3	0	0	1	1	0	1	1	0
4	0	1	0	0	0	1	1	1
5	0	1	0	1	1	0	0	0
6	0	1	1	0	1	0	0	1
7	0	1	1	1	1	0	1	0
8	1	0	0	0	1	0	1	1
9	1	0	0	1	1	1	0	0
10	1	0	1	0	X	X	X	X
11	1	0	1	1	X	X	X	X
12	1	1	0	0	X	X	X	X
13	1	1	0	1	X	X	X	X
14	1	1	1	0	X	X	X	X
15	1	1	1	1	X	X	X	X

- K-maps:

AB \ CD		C			
		00	01	11	10
A	00				
	01		1	1	1
	11	X	X	X	X
	10	1	1	X	X
		D			
		W			

AB \ CD		C			
		00	01	11	10
A	00		1	1	1
	01	1			
	11	X	X	X	X
	10		1	X	X
		D			
		X			

AB \ CD		C			
		00	01	11	10
A	00	1		1	
	01	1		1	
	11	X	X	X	X
	10	1		X	X
		D			
		Y			

AB \ CD		C			
		00	01	11	10
A	00	1			1
	01	1			1
	11	X	X	X	X
	10	1		X	X
		D			
		Z			

Example 2: BCD to Excess-3 Converter (Optimization)

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	1	1	1
11	X	X	X	X
10	1	1	X	X

W

AB \ CD	00	01	11	10
00	0	1	1	1
01	1	0	0	0
11	X	X	X	X
10	0	1	X	X

X

AB \ CD	00	01	11	10
00	1	0	1	0
01	1	0	1	0
11	X	X	X	X
10	1	0	X	X

Y

AB \ CD	00	01	11	10
00	1	0	0	1
01	1	0	0	1
11	X	X	X	X
10	1	0	X	X

Z

The final set of output equations are

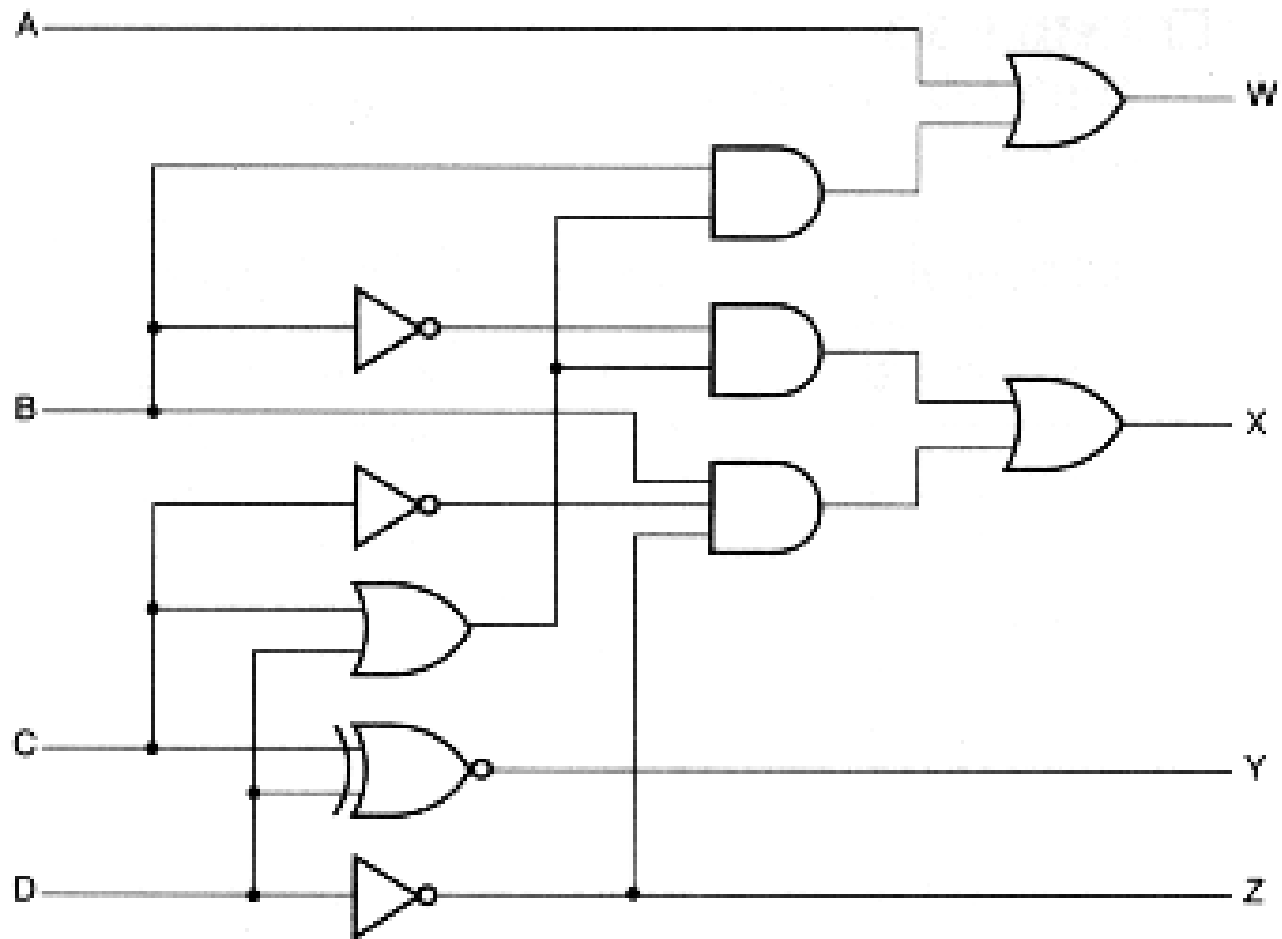
$$W = A + BC + BD = A + B(C + D)$$

$$X = B'C + B'D + BC'D' = B'(C + D) + BC'D'$$

$$Y = CD + C'D'$$

$$Z = D'$$

Logic Diagram



Three-Variable K-Maps

$$f = \sum(0,4) = \overline{B}\overline{C}$$

A \ BC	BC			
	00	01	11	10
0	1	0	0	0
1	1	0	0	0

$$f = \sum(4,5) = A\overline{B}$$

A \ BC	BC			
	00	01	11	10
0	0	0	0	0
1	1	1	0	0

$$f = \sum(0,1,4,5) = \overline{B}$$

A \ BC	BC			
	00	01	11	10
0	1	1	0	0
1	1	1	0	0

$$f = \sum(0,1,2,3) = \overline{A}$$

A \ BC	BC			
	00	01	11	10
0	1	1	1	1
1	0	0	0	0

$$f = \sum(0,4) = \overline{A}C$$

A \ BC	BC			
	00	01	11	10
0	0	1	1	0
1	0	0	0	0

$$f = \sum(4,6) = A\overline{C}$$

A \ BC	BC			
	00	01	11	10
0	0	0	0	0
1	1	0	0	1

$f = \sum(0,2) = \overline{A}\overline{C}$

	BC			
	00	01	11	10
A				
0	1	0	0	1
1	0	0	0	0

$f = \sum(0,2,4,6) = \overline{C}$

	BC			
	00	01	11	10
A				
0	1	0	0	1
1	1	0	0	1

AB \ CD	CD			
	00	01	11	10
00	1	0	0	0
01	0	0	0	0
11	0	0	0	0
10	1	0	0	0

$f = \sum (0,8) = \bar{B} \cdot \bar{C} \cdot \bar{D}$

AB \ CD	CD			
	00	01	11	10
00	0	0	0	0
01	0	1	0	0
11	0	1	0	0
10	0	0	0	0

$f = \sum (5,13) = B \cdot \bar{C} \cdot D$

AB \ CD	CD			
	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	0	1	1	0
10	0	0	0	0

$f = \sum (13,15) = A \cdot B \cdot D$

AB \ CD	CD			
	00	01	11	10
00	0	0	0	0
01	1	0	0	1
11	0	0	0	0
10	0	0	0	0

$f = \sum (4,6) = \bar{A} \cdot B \cdot \bar{D}$

AB \ CD	CD			
	00	01	11	10
00	0	0	1	1
01	0	0	1	1
11	0	0	0	0
10	0	0	0	0

$f = \sum (2,3,6,7) = \bar{A} \cdot C$

AB \ CD	CD			
	00	01	11	10
00	0	0	0	0
01	1	0	0	1
11	1	0	0	1
10	0	0	0	0

$f = \sum (4,6,12,14) = B \cdot \bar{D}$

AB \ CD	CD			
	00	01	11	10
00	0	0	1	1
01	0	0	0	0
11	0	0	0	0
10	0	0	1	1

$f = \sum(2,3,10,11) = \bar{B} \cdot C$

AB \ CD	CD			
	00	01	11	10
00	1	0	0	1
01	0	0	0	0
11	0	0	0	0
10	1	0	0	1

$f = \sum(0,2,8,10) = \bar{B} \cdot \bar{D}$

AB \ CD	CD			
	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	0	0	0	0
10	0	0	0	0

$f = \sum(4,5,6,7) = \bar{A} \cdot B$

AB \ CD	CD			
	00	01	11	10
00	0	0	1	0
01	0	0	1	0
11	0	0	1	0
10	0	0	1	0

$f = \sum(3,7,11,15) = C \cdot D$

AB \ CD	CD			
	00	01	11	10
00	1	0	1	0
01	0	1	0	1
11	1	0	1	0
10	0	1	0	1

$f = \sum(0,3,5,6,9,10,12,15)$
 $f = A \oplus B \oplus C \oplus D$

AB \ CD	CD			
	00	01	11	10
00	0	1	0	1
01	1	0	1	0
11	0	1	0	1
10	1	0	1	0

$f = \sum(1,2,4,7,8,11,13,14)$
 $f = A \oplus B \oplus C \oplus D$

		CD			
		00	01	11	10
AB	00	0	1	1	0
	01	0	1	1	0
	11	0	1	1	0
	10	0	1	1	0

$f = \sum(1,3,5,7,9,11,13,15)$
 $f = D$

		CD			
		00	01	11	10
AB	00	1	0	0	1
	01	1	0	0	1
	11	1	0	0	1
	10	1	0	0	1

$f = \sum(0,2,4,6,8,10,12,14)$
 $f = \bar{D}$

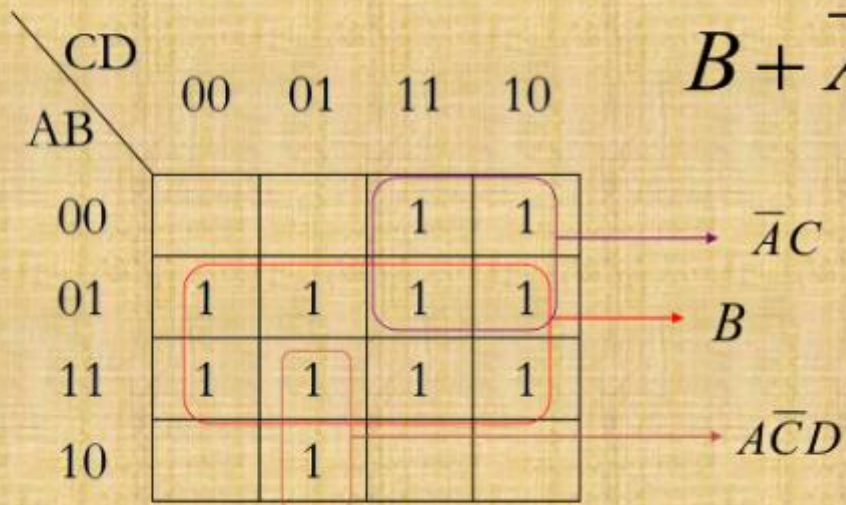
		CD			
		00	01	11	10
AB	00	0	0	0	0
	01	1	1	1	1
	11	1	1	1	1
	10	0	0	0	0

$f = \sum(4,5,6,7,12,13,14,15)$
 $f = B$

		CD			
		00	01	11	10
AB	00	1	1	1	1
	01	0	0	0	0
	11	0	0	0	0
	10	1	1	1	1

$f = \sum(0,1,2,3,8,9,10,11)$
 $f = \bar{B}$

Determining the Minimum SOP Expression from the Map



$$B + \bar{A}C + A\bar{C}D$$

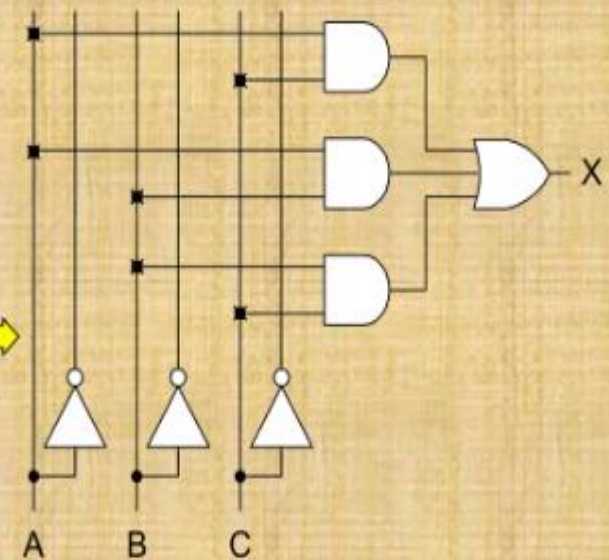
Example: Design a 3-input (A,B,C) digital circuit that will give at its output (X) a logic 1 only if the binary number formed at the input has more ones than zeros.

	Inputs			Output
	A	B	C	
0	0	0	0	0
1	0	0	1	0
2	0	1	0	0
3	0	1	1	1
4	1	0	0	0
5	1	0	1	1
6	1	1	0	1
7	1	1	1	1

$$X = \sum (3, 5, 6, 7)$$

A \ BC	00	01	11	10
0	0	0	1	0
1	0	1	1	1

$$X = AC + AB + BC$$



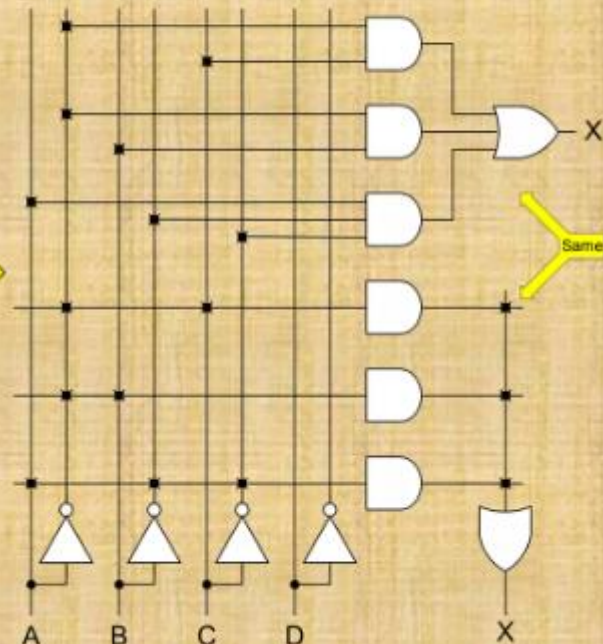
Example: Design a 4-input (A,B,C,D) digital circuit that will give at its output (X) a logic 1 only if the binary number formed at the input is between 2 and 9 (including).

	Inputs				Output
	A	B	C	D	
0	0	0	0	0	0
1	0	0	0	1	0
2	0	0	1	0	1
3	0	0	1	1	1
4	0	1	0	0	1
5	0	1	0	1	1
6	0	1	1	0	1
7	0	1	1	1	1
8	1	0	0	0	1
9	1	0	0	1	1
10	1	0	1	0	0
11	1	0	1	1	0
12	1	1	0	0	0
13	1	1	0	1	0
14	1	1	1	0	0
15	1	1	1	1	0

$$X = \sum (2,3,4,5,6,7,8,9)$$

CD \ AB	00	01	11	10
00	0	0	1	1
01	1	1	1	1
11	0	0	0	0
10	1	1	0	0

$$X = \overline{A}C + \overline{A}B + A\overline{B}\overline{C}$$



Find the implicants of $F(A, B, C, D) = \sum m(0, 2, 3, 6, 7, 8, 10, 12, 13)$ using QM method

Step 1: List all the minterms in their binary form. For each minterm, write the corresponding 4-bit binary representation.

Minterm	Binary Representation
m_0	0 0 0 0
m_2	0 0 1 0
m_3	0 0 1 1
m_6	0 1 1 0
m_7	0 1 1 1
m_8	1 0 0 0
m_{10}	1 0 1 0
m_{12}	1 1 0 0
m_{13}	1 1 0 1

Step 2: Arrange these binary minterms based on the number of '1's they contain. All minterms with the same count of '1's should be grouped.

	Minterm	Binary Representation	
	m_0	0 0 0 0	✓
	m_2	0 0 1 0	✓
	m_8	1 0 0 0	✓
	m_3	0 0 1 1	✓
	m_6	0 1 1 0	✓
	m_{10}	1 0 1 0	✓
	m_{12}	1 1 0 0	✓
	m_7	0 1 1 1	✓
	m_{13}	1 1 0 1	✓

Step 3:

Compare each binary number with every term in the adjacent next higher category and if they differ only by one position, put a check mark and copy the term into the next column with '–' in the position that they differed.

Minterms	Binary Representation	
0, 2	0 0 - 0	✓
0, 8	- 0 0 0	✓
2, 3	0 0 1 -	✓
2, 6	0 - 1 0	✓
2, 10	- 0 1 0	✓
8, 10	1 0 - 0	✓
8, 12	1 - 0 0	
3, 7	0 - 1 1	✓
6, 7	0 1 1 -	✓
12, 13	1 1 0 -	

Step 4:

Apply the same process described in step 3 for the resultant column and continue these cycles until a single pass through the cycle yields no further elimination of literals.

Minterms	Binary Representation
0, 2, 8, 10	- 0 - 0
2, 3, 6, 7	0 - 1 -

The implicants of the function

$$\bar{B}\bar{D}, \bar{A}C, A\bar{C}\bar{D} \text{ and } ABC$$

Determine the minterms for $f(a,b,c,d)=\sum m(0,1,2,5,6,7,8,9,10,14)$ using QM method

Step 1

group 0	<u>0</u>	<u>0000</u>
group 1	{	1 0001
		2 0010
		<u>8 1000</u>
group 2	{	5 0101
		6 0110
		9 1001
		<u>10 1010</u>
group 3	{	7 0111
		<u>14 1110</u>

Step 2

group 0	<u>0 0000</u>	✓
group 1	1 0001	✓
	2 0010	✓
	<u>8 1000</u>	✓
group 2	5 0101	✓
	6 0110	✓
	9 1001	✓
	<u>10 1010</u>	✓
group 3	7 0111	✓
	<u>14 1110</u>	✓

Step 3

0, 1	000–	✓
0, 2	00–0	✓
0, 8	–000	✓
1, 5	0–01	
1, 9	–001	✓
2, 6	0–10	✓
2, 10	–010	✓
8, 9	100–	✓
8, 10	10–0	✓
5, 7	01–1	
6, 7	011–	
6, 14	–110	✓
10, 14	1–10	✓

Step 4

0, 1, 8, 9	00
0, 2, 8, 10	00
0, 8, 1, 9	00
0, 8, 2, 10	00
2, 6, 10, 14	-- 10
2, 10, 6, 14	-- 10

Implicants of the function are

$$c\bar{d}, \bar{b}\bar{d}, \bar{b}\bar{c}, \bar{a}\bar{c}d, \bar{a}bd, \bar{a}bc$$

THANK YOU